

ULTRA-WIDEBAND CONFORMAL HELMET ANTENNA

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This paper presents the development of an ultra-wideband (300 to 3000 MHz), vertically polarized, nearly omni-directional (in azimuth) communications antenna integrated into the camouflage cover of a standard military-issue Kevlar helmet. The Helmet Camouflage Cover Antenna (referred to as the "helmet antenna") is one of three antennas (with the combined frequency coverage from 2 to 2000 MHz) based on the antenna COMbat Wear INtegration (COMWIN) concept developed at the Naval Postgraduate School (NPS) for the man-portable implementation of the new Joint Tactical Radio System (JTRS). The results of computer simulations and prototype measurements for the helmet antenna are presented. The helmet antenna is conformal to the shape of the Kevlar ballistic helmet. It is constructed of conducting cloth (polyester interwoven with nickel/copper fibers) and attached as a liner to the helmet's camouflage cover. The helmet antenna is therefore visually undetectable, except for the connector in the back of the helmet. Several variants of the helmet antenna have been designed and simulated at NPS using the High Frequency Structure Simulator (HFSS) software, with different trade-offs between the low frequency cut-off and the maximum VSWR "ripple" within the operational band. Two prototypes, Mk I [1] and Mk II, have been built and measured at the NPS. Measurements of the input impedance/VSWR as a function of frequency for the Mk II prototype show performance exceeding the HFSS computer model predictions. The VSWR is less than 3:1 for all frequencies between 300 and 3000 MHz. The presence of an operator wearing the helmet does not affect the VSWR significantly.

1 Introduction

The Joint Tactical Radio System (JTRS) is being developed in the US to meet the emerging needs for ultra-wideband tactical radios. The desired 2 MHz to 2000 MHz operating frequency range poses a major challenge with regard to the antenna system, especially for the man-portable implementation. The conventional "whip" antennas typically used for man-portable tactical radios are incapable of supporting such a large bandwidth. In addition, "whip" antennas also have significant visual profiles that expose the operator's position. This paper presents the results of computer simulation and measurements of the conformal helmet antenna (Mk II) prototype. The measured VSWR (relative to 50 Ω) for the Mk II prototype is less than 3 for all frequencies between 300 and 3000 MHz.

2 The Helmet Antenna Design

The antenna is comprised of two radiating surfaces separated by a gap. The top and bottom radiating surfaces have approximately equal areas for the optimum electrical performance (maximum bandwidth). The conducting strip in the front of the helmet provides connection between the top and the bottom half of the antenna. A coaxial feed line at the back of the helmet feeds the antenna (across the gap). The inner conductor of the coaxial feed is soldered to the upper half of the antenna while the outer conductor (shield) is soldered to the lower half. The size and geometry of the gap separating the top and the bottom halves control the value of the input impedance and the antenna bandwidth

(defined by the condition $VSWR < 3$ for 50Ω reference). The gap orientation and the feed arrangement create a vertically polarized far field, whereas the helmet shape creates a nearly omnidirectional radiation in azimuth. The conducting cloth that makes up the radiating surfaces is made of polyester interwoven with nickel/copper fiber, and has the surface resistivity of less than $0.1 \Omega/\text{square}$, the specific weight of less than 3.3 oz/yd^2 , and the thickness of only 0.006 inches. These properties allow the antenna to be efficient, lightweight, and conformal to the helmet. The helmet antenna Mk II prototype is shown in Figure 1. The Mk II has a gap that is 0.5 cm wide at the front, gradually increases to about 8.5 cm on the side and then decreases to 0.5 cm at the rear of the helmet, where the coaxial feed line is attached.



Figure 1. Mk II Prototype Helmet Antenna

3 Comparison of HFSS Simulation and Prototype Measurements

Computer simulations of the helmet antenna were performed using Ansoft's HFSS finite element software. The simulated and measured VSWR (50Ω reference) for the Mk II prototype antenna are shown in Figure 2. The measured VSWR is generally lower than the predicted VSWR, especially below 600 MHz and above 1200 MHz.

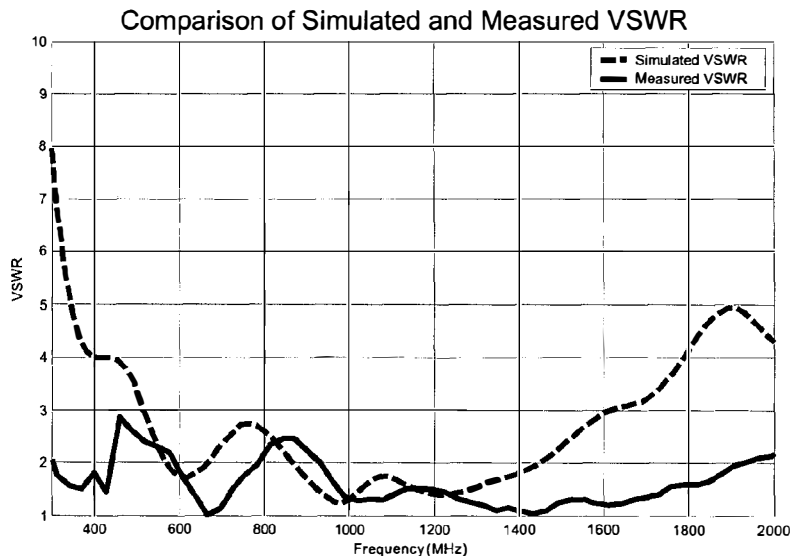


Figure 2. Comparison of Simulated and Measured VSWR

4 Measurement Results

The input impedance measurements from 45 to 3000 MHz were completed using the Hewlett Packard Vector Network Analyzer HP 8510. The real and imaginary parts of the helmet antenna input impedance were measured for the antenna without the helmet, for the antenna mounted on the helmet, and for the antenna on the helmet worn by one of the authors. The same 50 Ω feed line was used for all measurements. The VSWR for each measurement was calculated and plotted. The measured VSWR for the Mk II prototype antenna is less than 3 from about 280 MHz to above 3000 MHz.

4.1 Effect of Kevlar Helmet on the Helmet Antenna VSWR

The measured VSWR for the Mk II prototype antenna with and without Kevlar helmet is shown in Figure 3. Mounting the antenna on the helmet improves the VSWR at frequencies below 2000 MHz, which are also the frequencies that the helmet antenna is designed to operate at. The improvement in the VSWR is primarily due to “dielectric loading” of the antenna by the Kevlar material, as indicated by the VSWR curve “shift” towards lower frequencies (with the antenna mounted on the helmet).

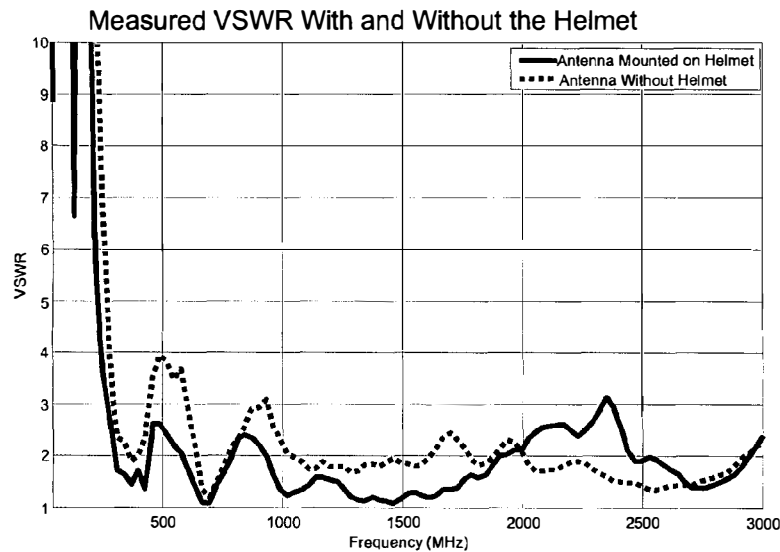


Figure 3. Comparison of Measured VSWR With and Without the Helmet

4.2 Effect of a Person Wearing Kevlar Helmet on the Helmet Antenna VSWR

The measured VSWR for the helmet-mounted Mk II prototype antenna with and without an operator wearing the helmet is plotted in Figure 4. The presence of the operator improves the VSWR slightly for frequencies below 2000 MHz. The improvement in the VSWR is due to a combination of dielectric loading and resistive loss of the operator wearing the helmet, the latter being undesirable from the radiation hazard standpoint and the antenna radiation efficiency reduction (when RF power is absorbed by the operator rather than radiated). However, the change in the VSWR is relatively small, indicating that the Kevlar helmet “decouples” the operator from the antenna, thus reducing the operator’s risk of RF radiation exposure.

5 Advantages of the Conformal Helmet Antenna

Compared to the conventional antennas, the helmet antenna has the following main advantages:

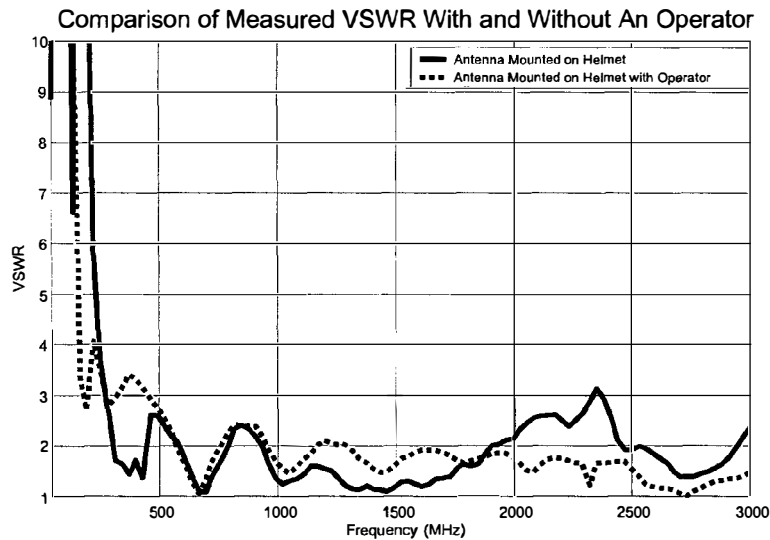


Figure 4. Comparison of Measured VSWR With and Without an Operator Wearing the Helmet

- Efficient operation from 300 MHz to 3,000 MHz frequency range without an antenna tuner (ultra-wide instantaneous bandwidth)
- No visual signature (radio operator indistinguishable from any other soldier)
- No set-up required for the antenna
- No damage to the antenna from foliage and/or obstacles
- All-weather antenna operation
- Light weight
- Inexpensive to manufacture and maintain

This unique combination of features, to our knowledge, no conventional man-portable antenna can provide.

6 Conclusions

The authors have designed, built, and tested an ultra-wideband, lightweight, conformal helmet antenna that operates from 300 MHz to above 3000 MHz (a bandwidth ratio exceeding 10:1) with the VSWR < 3 . The antenna is superior to conventional man-portable communications antennas in terms of performance, cost, and maintenance. Of particular interest to military applications is the absence of any visual signature. The helmet antenna development at the NPS is continuing, while an independent laboratory has been tasked to verify that the antenna meets the RF safety requirements.

Acknowledgements

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References

- [1] A.T. Tan, "Design of an Ultra-Wideband Low Profile Vertically Polarized UHF Antenna for the

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